

Activating - 2

① A CCTV image contains salt-and-pepper noise due to transmission error.

Technique used: Median Filter

Reason: Salt-and-pepper noise appears as random black and white pixels due to transmission error or faulty.

② A satellite image is corrupted by Gaussian noise from sensor.

Technique used: Gaussian Filter (Gaussian Blur)

Reason: The Gaussian Filter smooths the image and effectively reduces Gaussian noise caused by imaging sensor while maintaining overall image quality.

③ A medical x-ray image has random noise but fine edges must be preserved.

Technique used: Bilateral Filter

Reason: The Bilateral Filter removes random noise while preserving important edges, making it suitable for medical images where fine details are essential.

④ A scanned document has black and white dots scattered randomly.

Technique used: Median Filter

Reason: The median filter removes scattered black and white dots without affecting the text (or) document edges.

④ A Scanned doc A low light image show heavy grain without clear boundaries.
Technique used: Non-Local means Denoising
Reason: This technique effectively removes heavy grain noise while preserving image details and textures better than traditional filters.

⑤ A foggy road image has very low contrast.
Technique used: Histogram Equalization
Reason: Histogram Equalization improves the image contrast, making roads and objects more visible in foggy conditions.

⑥ An underwater image appears dull and washed out.
Technique used: CLAHE
Reason: CLAHE enhances local contrast and improves the visibility of underwater objects without amplifying noise excessively.

⑦ A Chest X-ray shows poor visibility of internal structures.
Technique used: CLAHE
Reason: CLAHE increases local contrast, making internal organs and tissues more visible while preserving important medical details.

⑧ A night-time traffic image lacks intensity.
Technique used: Histogram Equalization
Reason: Histogram Equalization improves brightness and contrast, making vehicles and roads easier to identify in low-light conditions.

⑩ - A grayscale image uses only a narrow intensifying range.

Technique used: ~~Contrast stretching~~

Reason: Contrast ~~stretching~~ expands the narrow intensifying range to use the full grayscale spectrum, resulting in a clearer and higher-contrast image.

